



Antimicrobial Peptides

A Roadmap for Accelerating Discovery and Development

2025, Pages xiii-xiv

Introduction—Antimicrobial peptides: From the bench to the bedside

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Available online 17 January 2025, Version of Record 17 January 2025.

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<https://doi.org/10.1016/B978-0-443-15393-8.10000-4> ↗

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Over the centuries, humanity has continuously waged war against infections, diseases, and illnesses, predominantly caused by various microbes (Ventola, 2015). This perpetual struggle culminated in the discovery of penicillin by Sir Alexander Fleming in 1928, marking a pivotal advancement in medical science. Penicillin represented an innovative approach to combating microbial infections, significantly contributing to the preservation of innumerable lives in the ensuing years (Dadgostar et al., 2019; Prestinaci et al., 2015;

[Ventola, 2015](#)). However, the advent of antimicrobial drugs also introduced a formidable public health challenge: antimicrobial resistance (AMR) ([Ventola, 2015](#)). AMR is characterized by a natural diminution in the susceptibility of microbes to therapeutic agents over time, enabling these organisms to proliferate despite the presence of medications that were previously effective. ([Dadgostar et al., 2019](#); [Prestinaci et al., 2015](#)). Therefore, infection with microbial-resistant organisms is considered life-threatening, as it results in serious illness, increased medical and healthcare costs, increased morbidity, and even increased mortality ([Dadgostar et al., 2019](#); [Prestinaci et al., 2015](#)). Consequently, infections caused by resistant organisms pose severe health risks, leading to grave illnesses, escalated medical and healthcare expenses, increased morbidity, and heightened mortality rates. The Center for Disease Control and Prevention (CDC) reports that AMR is responsible for the deaths of at least 1.27 million individuals worldwide and was directly linked to 5 million fatalities in 2019 alone ([National Infection and Death Estimates for Antimicrobial Resistance, 2021/2021](#)). Furthermore, the economic burden of AMR in the United States is estimated to reach approximately \$55 billion annually, attributed to increased healthcare costs and lost productivity ([Dadgostar et al., 2019](#); [National Infection and Death Estimates for Antimicrobial Resistance, 2021/2021](#); [Prestinaci et al., 2015](#)).

How is AMR generated? The genesis of AMR is multifaceted, yet the principal catalyst is the overexposure of microbes to pharmacological agents. This exposure fosters natural or intrinsic genetic alterations within the microorganisms, enabling them to adapt to the consistently adverse environments they encounter ([Dadgostar et al., 2019](#); [Prestinaci et al., 2015](#); [Ventola, 2015](#)). More precisely, the predominant factors contributing to AMR include the excessive use of antimicrobials, improper administration of these medications by patients, their application in agricultural and farming practices, and regulatory obstacles that impede the advancement of new therapeutic agents ([Dadgostar et al., 2019](#); [Prestinaci et al., 2015](#); [Ventola, 2015](#)). These factors are particularly critical concerning antibacterial resistance. Bacteria may develop resistance through spontaneous evolution, acquisition from their own species, horizontal gene transfer from nonrelated entities through plasmids, and simple genetic mutations ([Dadgostar et al., 2019](#); [Prestinaci et al., 2015](#); [Ventola, 2015](#)). Consequently, acknowledging the urgent need to combat antimicrobial-resistant microorganisms, there has been a continuous effort to innovate and explore new technologies aimed at addressing this global challenge.

Among various innovative strategies proposed to mitigate AMR, noteworthy methods include bacteriophage therapy, immunotherapy, CRISPR/Cas interventions, fecal microbiota transplantation, combination drug therapies, and the utilization of antibiotic adjuvants targeting bacterial nonessential functions. Notably, one of the most promising alternatives

in this arsenal is the development of antimicrobial peptides (AMPs) ([Annunziato, 2019](#); [Huan et al., 2020](#); [Kaur, 2016](#); [Qadri et al., 2021](#)). AMPs are characterized as a distinct group of peptides, generally comprising tens to hundreds of amino acids, synthesized by a wide array of organisms as a natural defense mechanism against a multitude of microbes, including bacteria, fungi, and viruses ([Bahar & Ren, 2013](#); [Huan et al., 2020](#)). These peptides offer considerable benefits compared with traditional therapeutic approaches due to their high specificity, natural abundance, ease of synthesis and replication through gene libraries and recombinant technologies, and broad antimicrobial efficacy ([Bahar & Ren, 2013](#); [Hancock & Chapple, 1999](#); [Huan et al., 2020](#); [Lei et al., 2019](#)). Since Alexander Fleming's discovery of lysozyme in 1922, research on AMPs has significantly expanded ([Hancock & Chapple, 1999](#); [Huan et al., 2020](#)). Pioneers in this field, such as René Dubos, who extracted an antimicrobial agent from a *Bacillus* strain to combat pneumococcal infections in mice in 1939, and Hans Boman, who delineated the structure of innate insect AMPs in 1981, have highlighted the potential of leveraging nature's own mechanisms to counteract microbial threats ([Bahar & Ren, 2013](#); [Nakatsuji & Gallo, 2012](#)).

Consequently, recognizing their pivotal role in contemporary science and the battle against AMR, an in-depth grasp of the latest progress in the domain of AMPs is essential for devising innovative strategies for their discovery, utilization, and synthesis. Hence, this book serves as an exhaustive compilation of both computational and experimental methodologies for the identification, production, and clinical application of AMPs. It is structured into three comprehensive sections, each dedicated to exploring the forefront of AMP research. Specifically, Section A delves into computational techniques for AMP discovery, Section B discusses experimental methods pertinent to AMPs, and Section C addresses the translational research in the field of AMPs, offering insights into their practical applications and potential impact on healthcare.

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